Tyler Mitts

1+ 417-379-7404 • tmitts0206@gmail.com • tylermitts.com •

Education

University of Missouri - Columbia
Bachelor of Science in Computer Science (GPA: 3.6/4.0) Minor in Mathematics and IT

Projects

WHERETOSTARGAZE.us (1st Place, TigerHacks 2025)
Technologies: React, FastAPI, AWS S3, Redis, LeafletJS, OpenRoute API

- Engineered a web app to find optimal stargazing locations by analyzing light pollution, cloud coverage, and forest density data
- Implemented Redis caching to minimize API costs and reduce load times by 40%
- Built interactive map visualizations with LeafletJS using OpenRoute isochrones and distance heatmaps
- Developed FastAPI backend with AWS S3 for large dataset storage and deployed full application on Vercel

Depth Estimation for Minecraft Schematic Conversion
Technologies: Python, PyTorch, OpenCV, NumPy, scikit-learn

- Implemented MiDaS neural network for monocular depth estimation to reconstruct images in 3D
- Applied K-means clustering to segment depth maps into distinct layers for z-coordinate mapping, using elbow method for optimal k value
- Developed nearest-neighbor algorithm to match RGB values to Minecraft block textures for accurate in-game recreation

Cancer Cell Segmentation and Analysis
Technologies: Python, OpenCV, scikit-learn, NumPy

- Implemented multi-stage segmentation pipeline using K-means clustering, distance transform, and watershed algorithms to isolate individual cells
- Applied post-processing morphological operations (erosion, dilation) to refine segmentation boundaries and improve accuracy
- Utilized connected component analysis to count cells and compute area measurements, enabling quantitative cellular analysis

Experience

University of Missouri Division of Plant Science and Technology September 2025 – Current
TDA Research Assistant

- Applied topological data analysis (TDA) techniques including Vietoris-Rips complexes and level-set methods to extract shape information from plant cell data
- Implemented statistical analysis using Wasserstein and bottleneck distances with MDS and PCA to compare real and synthetic cellular structures

Mitts Technology LLC November 2025 – Current
Full-Stack Developer - ApplyApply.org

- Developed a full-stack job application automation platform with React frontend, AWS serverless backend
- Integrated AI-powered document generation using AWS Bedrock, Stripe payment processing, and job search API
- Implemented rate limiting, caching strategies, and Firebase authentication to support subscription-based SaaS model

City of Willard Pool June 2022 – August 2025
Concessions Manager

- Supervised and trained concession staff, fostering teamwork and enhancing leadership and management abilities
- Managed inventory and upkeep of equipment within concessions, as well as maintaining health regulations

University of Missouri Honors College August 2023 – Current
Front Desk

- Supported college events by preparing materials, welcoming guests, and managing phone calls